

- **Project Part 2: Database Requirements due Sunday, Sep 27**
 - **Introduction:** The Library database management system will be a robust and easy to use tool to keep track of the transactions and inventory of books for both library customers and the staff at a library. The database system will keep track of inventory, allow users to sign in and manage their account, check out and return books, as well as pay fees. The staff can see and manage users accounts, check books in, add books, and miscellaneous administrative tasks that may need to be used to manage the traffic and inventory of books.
 - **Stakeholders:** Library Users, Librarians, Inventory managers, System Administrators, IT
 - **Hardware and Software Requirements**
 - Database Engine: MySQL
 - Staff Computers, User Computers, Server Storage
 - **Project Requirements**
 - Functions for accessing/updating the database
 - Data entity descriptions
- **Data tables**
 - **Books/digital media items**
 - **Attributes:** title, author, ISBN, publication year, genre, and availability status (Returned, Due, Overdue). Special borrowing restrictions (Specific amounts of books that can be borrowed (5?))
 - **Relationships:** a book has a current holder (client), a transaction holds an item (book).
 - **Magazines**
 - **Attributes:** title, issue number, publication date, and availability status.
 - **Relationships:** a magazine has a current holder (client), a transaction holds an item (magazine).
 - **Clients**
 - **Attributes:** unique ID, name, email, phone number, membership type, and account status. Also fees accrued.
 - **Membership types:** regular, student, senior, maybe more.
 - These types have their own fee rates.
 - **Relationships:** a transaction has a creator/closer (client).
 - **Transactions**
 - **Attributes:** state (reserved, canceled, borrowed, returned), client, book/magazine, dates (borrowed, returned, due)
 - **Atomic Transaction:** Need to make sure that upon a successful checkout, a book is removed from available Books, and on a failed checkout, a book is not removed from available books
- **Constraints**

- **Time:** We have limited time and must remain tightly coupled to scope to ensure scope does not exceed the time we have allotted
 - Rare Book/Magazine Borrowing Restrictions
 - Maximum books checked out based on membership status
 - Fee Calculations on overdue books
- **Notification system**
 - **Customer:** Upcoming due dates, overdue items, available reserved items, held items
 - **Staff:** Overdue notification, books returned and checked out
- **Report queries**
 - Calculations of fines, book availability, borrowing trends, and client activity.
- **Front-end interface**
 - **Library staff:** check out items, process returns, add new items, and manage client accounts.
 - **Clients:** search the catalog, reserve items, check loan status, and manage fees.